

Problem Set for 24 March 2025

Exercise 0 Explain (without proving) the *implicit expressions*.

1. $\ker g$ as a subspace of $\ker fg$.

Solution. The linear map

$$\ker g \rightarrow \ker fg, \quad [x \mapsto x].$$

It is well-defined and injective.

2. $\operatorname{im}(fg)$ as a quotient space of $\operatorname{im}(g)$.
3. $\operatorname{Hom}(V, W) \times \operatorname{Hom}(U, V) \rightarrow \operatorname{Hom}(U, W)$ the bi-linear map of composition

Solution. The bi-linear map

$$\operatorname{Hom}(V, W) \times \operatorname{Hom}(U, V) \rightarrow \operatorname{Hom}(U, W), \quad (g, f) \mapsto g \circ f.$$

4. $\operatorname{Hom}(U, V) \rightarrow \operatorname{Hom}(U, W)$ induced by $f : V \rightarrow W$.
5. $\operatorname{Hom}(W, U) \rightarrow \operatorname{Hom}(V, U)$ induced by $f : V \rightarrow W$.
6. $\operatorname{Hom}_{\text{Sets}}(S, V) = V^n$ for finite set $|S| = n$.
7. $\mathbb{F}[x]$ as a subspace of $\operatorname{Hom}_{\mathbb{F}}(\mathbb{F}[[x]], \mathbb{F})$.
8. The $\mathbb{C}$ -linear space $\operatorname{Hom}_{\mathbb{R}}(\mathbb{C}, V)$, provided an $\mathbb{R}$ -linear space V .
9. **(Optional)** The set of all subsets of X forms a $\mathbb{F}_2$ -linear space, where $\mathbb{F}_2$ is a field with 2 elements.

Example Construct a *satisfying* linear isomorphism

$$\mathrm{Hom}_{\mathbb{F}}(U \times V, W) \simeq \mathrm{Hom}_{\mathbb{F}}(U, W) \times \mathrm{Hom}_{\mathbb{F}}(V, W).$$

Proof.

Step 1 Write down explicitly the linear map

$$\begin{aligned} \Phi : \quad \mathrm{Hom}_{\mathbb{F}}(U, W) \times \mathrm{Hom}_{\mathbb{F}}(V, W) &\rightarrow \mathrm{Hom}_{\mathbb{F}}(U \times V, W) \\ (f, g) &\mapsto [(u, v) \mapsto f(u) + g(v)]. \end{aligned}$$

For a very detailed expression,

$$\begin{aligned} \Phi : \quad \mathrm{Hom}_{\mathbb{F}}(U, W) \times \mathrm{Hom}_{\mathbb{F}}(V, W) &\rightarrow \mathrm{Hom}_{\mathbb{F}}(U \times V, W) \\ \left(\begin{bmatrix} U & \xrightarrow{f} & W \\ u & \mapsto & f(u) \end{bmatrix}, \begin{bmatrix} V & \xrightarrow{g} & W \\ v & \mapsto & g(v) \end{bmatrix} \right) &\mapsto \begin{bmatrix} U \times V & \xrightarrow{\Phi(f,g)} & W \\ (u, v) & \mapsto & f(u) + g(v) \end{bmatrix}. \end{aligned}$$

Step 2 Check that Φ is a well-defined linear map, i.e.

$$\Phi((f, g) + \lambda(f', g')) = \Phi(f, g) + \lambda\Phi(f', g'),$$

for both sides map (u, v) to $f(u) + f'(u) + \lambda g(v) + \lambda g'(v)$.

Step 3 Check that Φ is an injection. If $\Phi(f, g) = 0$, then so are the compositions

$$U \hookrightarrow U \times V \xrightarrow{\Phi(f,g)} W, \quad u \mapsto (u, 0) \mapsto f(u),$$

and

$$V \hookrightarrow U \times V \xrightarrow{\Phi(f,g)} W, \quad v \mapsto (0, v) \mapsto g(v).$$

It yields that $f = 0$ and $g = 0$.

Step 4 Check that Φ is a surjection. For any $\alpha : U \times V \rightarrow W$, consider

$$f_{\alpha} : U \rightarrow W, \quad u \mapsto \alpha(u, 0),$$

and

$$g_{\alpha} : V \rightarrow W, \quad v \mapsto \alpha(0, v).$$

Now $\Phi(f_{\alpha}, g_{\alpha}) = \alpha$, yielding that (f_{α}, g_{α}) is a preimage.

Step 5 Since Φ is well-defined, injective, and surjective, Φ is a linear isomorphism.

Your proof must include Step1-5.

Exercise 1 (optional if you have done this before) Show that $U \simeq \text{Hom}_{\mathbb{F}}(\mathbb{F}, U)$ for any $\mathbb{F}$ -linear space U .

Step 1 Write down explicitly the linear map $u \mapsto [1_{\mathbb{F}} \mapsto u]$, to be explicit,

$$\Phi : \quad \text{Big Formula...}$$

Step 2 Check that Φ is a well-defined linear map, i.e.,

Step 3 Show that Φ is injective, your solution...

Step 4 Show that Φ is surjective, your solution...

Step 5 Conclude that Φ is an linear isomorphism.

Exercise 2 (optional if you have done this before) Show that $U \simeq \operatorname{Hom}_{\mathbb{F}}(\operatorname{Hom}_{\mathbb{F}}(U, \mathbb{F}), \mathbb{F})$ if $\dim U < \infty$.

Step 1 Write down explicitly the linear map $u \mapsto [f \mapsto f(u)]$, to be explicit,

$$\Phi : \quad \text{Big Formula...}$$

Step 2 Check that Φ is a well-defined linear map, i.e.,

Step 3 Show that Φ is injective, your solution...

Step 4 Show that $\dim U = \dim \operatorname{Hom}_{\mathbb{F}}(U, \operatorname{Hom}_{\mathbb{F}}(U, \mathbb{F}))$ by induction.

Step 5 Conclude that Φ is a linear isomorphism by dimensional analysis.

Exercise 3 Let V be a linear space and $S \subset V$ is linearly independent (S is not necessary finite). Show that

$$\mathrm{Hom}_{\mathbb{F}}(\mathrm{Span}(S), \mathbb{F}) \simeq \mathrm{Hom}_{\mathrm{Sets}}(S, \mathbb{F}).$$

Step 1 Write down explicitly the linear map $f \mapsto (f(s))_{s \in S}$, to be explicit,

$$\Phi : \quad \text{Big Formula...}$$

Step 2 Check that Φ is a well-defined linear map, i.e.,

Step 3 Show that Φ is injective, your solution...

Step 4 Show that Φ is surjective, your solution...

Step 5 Conclude that Φ is an linear isomorphism.

In particular, $\mathbb{F}[[x]] \simeq \mathrm{Hom}_{\mathbb{F}}(\mathbb{F}[x], \mathbb{F})$.

Exercise 4 Recall that $\mathbb{C}$ -linear spaces are automatically $\mathbb{R}$ -linear, and so are $\mathbb{C}$ -linear maps. Show that

$$\mathrm{Hom}_{\mathbb{R}}(U, V) \simeq (\mathrm{Hom}_{\mathbb{C}}(U, V))^2 \quad (\text{as } \mathbb{R}\text{-linear spaces}).$$

Here $W^2 := W \times W$.

Step 0 Show that $\mathrm{Hom}_{\mathbb{C}}(U, V)$ is indeed an $\mathbb{R}$ -linear space.

Step 1 Write down explicitly the linear map $f \mapsto (f_1, f_2)$, to be explicit,

$$\Phi : \quad \text{Big Formula...}$$

Step 2 Check that Φ is a well-defined linear map, i.e.,

Step 3 Show that Φ is injective, your solution...

Step 4 Show that Φ is surjective, your solution...

Step 5 Conclude that Φ is an linear isomorphism.

(Optional Exercise) Determine

$$\mathrm{Hom}_{\mathbb{R}}(U, V) \stackrel{?}{\simeq} (\mathrm{Hom}_{\mathbb{C}}(U, V))^2 \quad (\text{as } \mathbb{C}\text{-linear spaces}).$$